# Excretion in humans

20 questions: 10 multiple choice, 10 structured.

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## 1 · Part 1 Core Excretion in humans [1]

Which organ excretes carbon dioxide from the human body?

A kidneys

B lungs

C liver

D stomach

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## 2 · Part 1 Core Excretion in humans [1]

Which row shows the organ that excretes carbon dioxide and the organ that excretes urea?

| | **carbon dioxide** | **urea** |
|---|---|---|
| A | kidneys | lungs |
| B | liver | kidneys |
| C | lungs | liver |
| D | lungs | kidneys |

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## 3 · Part 1 Core Excretion in humans [1]

Which list contains only substances that are excreted by the kidneys?

A urea, excess water and excess ions

B urea, glucose and carbon dioxide

C carbon dioxide, excess water and proteins

D glucose, excess ions and amino acids

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## 4 · Part 1 Core Excretion in humans [1]

One afternoon, a student drinks 2 $dm^3$ of water in a short time. Over the next two hours, the student produces a much larger volume of urine than usual.

Which statement explains this?

A the extra water is converted into urea in the liver

B the kidneys stop excreting urea

C the kidneys excrete the excess water from the blood

D the bladder absorbs water from the urine

## 5 · Part 1 Supplement Excretion in humans [1]

What is deamination?

A the removal of the nitrogen-containing part of amino acids to form urea

B the joining together of amino acids to form proteins

C the breakdown of proteins into amino acids by protease

D the filtration of urea out of the blood in the kidneys

## 6 · Part 1 Supplement Excretion in humans [1]

The filtrate formed in the glomerulus flows along the nephron.

Which statement about the water in the filtrate of a healthy person is correct?

A all of the water in the filtrate is reabsorbed into the blood

B none of the water in the filtrate is reabsorbed into the blood

C the water in the filtrate is reabsorbed into the blood in the glomerulus

D most of the water in the filtrate is reabsorbed into the blood

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## 7 · Part 1 Supplement Excretion in humans [1]

Amino acids absorbed from the small intestine are carried to the liver. Liver cells use some of these amino acids to make proteins.

What is the name of this use of amino acids by the liver?

A deamination

B assimilation

C digestion

D egestion

## 8 · Part 1 Supplement Excretion in humans [1]

One person eats a meal that contains much more protein than the body needs. The protein is digested and the amino acids are absorbed into the blood.

What happens to the excess amino acids?

A they are stored in the kidneys and released when needed

B they are egested from the body in the faeces

C they are deaminated in the liver and urea is formed

D they are excreted by the lungs without being changed

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## 9 · Part 1 Supplement Excretion in humans [1]

In a healthy person, there is no glucose in the urine.

One person has a very high concentration of glucose in the blood. Glucose is found in this person's urine.

What is the most likely explanation?

A glucose is made from urea in the nephrons

B glucose is not filtered out of the blood at the glomerulus

C the nephrons reabsorb urea into the blood instead of glucose

D more glucose is filtered at the glomerulus than the nephrons can reabsorb

## 10 · Part 1 Supplement Excretion in humans [1]

Why is it important that urea is excreted from the body?

A urea is toxic, so it would harm the body if it built up in the blood

B urea is needed by the kidneys as their main source of energy

C urea would be converted back into amino acids if it stayed in the body

D urea is used by the liver to make new proteins

## 11 · Part 1 Core Excretion in humans [5]

The human body makes waste substances that must be excreted.

(a) State the organ that excretes carbon dioxide. [1]

(b) Carbon dioxide is made by a process that happens in all living cells. Name this process. [1]

(c) State three substances that are excreted by the kidneys. [3]

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## 12 · Part 1 Core Excretion in humans [6]

One student measures the total volume of liquid she drinks and the total volume of urine she produces on three different days. She eats similar food on all three days.

On day 3 she goes for a long run on a hot afternoon.

The table shows her results.

| day | volume of liquid drunk / $cm^3$ | volume of urine produced / $cm^3$ |
|---|---|---|
| 1 | 1500 | 1200 |
| 2 | 2500 | 2100 |
| 3 | 1500 | 600 |

(a) Use the results for day 1 and day 2 to describe the effect of drinking more liquid on the volume of urine produced. [1]

(b) Calculate the percentage increase in the volume of urine produced from day 1 to day 2. [2]

(c) The student drinks the same volume of liquid on day 1 and day 3. Suggest why she produces less urine on day 3. [2]

(d) Urine contains a waste substance that contains nitrogen. Name this substance. [1]

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## 13 · Part 1 Core Excretion in humans [5]

Scientists measure the concentrations of carbon dioxide and urea in the blood flowing into and out of two organs, P and Q. One of the organs is a lung and the other is a kidney.

The table shows the results.

| substance | blood entering P / arbitrary units | blood leaving P / arbitrary units | blood entering Q / arbitrary units | blood leaving Q / arbitrary units |
|---|---|---|---|---|
| carbon dioxide | 52 | 48 | 48 | 52 |
| urea | 0.30 | 0.30 | 0.30 | 0.24 |

(a) Use the data to identify organ P and organ Q. [2]

(b) The concentration of carbon dioxide is higher in the blood leaving organ Q than in the blood entering organ Q. Explain why. [2]

(c) Explain why the concentration of urea does not change as blood flows through organ P. [1]

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## 14 · Part 1 Supplement Excretion in humans [5]

The liver has important roles in dealing with amino acids.

(a) Describe the role of the liver in the assimilation of amino acids. [1]

(b) Describe what is meant by deamination. [2]

(c) Name the organ in which urea is formed. [1]

(d) Name the organ that excretes urea from the body. [1]

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## 15 · Part 1 Supplement Excretion in humans [5]

Each kidney contains about one million nephrons. At the start of each nephron, blood flows through a glomerulus, which is a small knot of blood capillaries. Some substances leave the blood in the glomerulus and enter the nephron, forming a liquid called filtrate. The filtrate then flows along the rest of the nephron, and the liquid that leaves the end of the nephron is urine.

(a) Name the process by which substances leave the blood in the glomerulus. [1]

(b) State four substances that leave the blood in the glomerulus and enter the nephron. [2]

(c) Glucose is present in the filtrate but is not normally present in urine. Explain why. [2]

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## 16 · Part 1 Supplement Excretion in humans [6]

The table shows the concentration of urea in three liquids from a healthy person.

| liquid | concentration of urea / g per $dm^3$ |
|---|---|
| blood plasma entering the glomerulus | 0.3 |
| filtrate in the nephron, just after the glomerulus | 0.3 |
| urine | 18.0 |

(a) State the evidence in the table that urea is filtered out of the blood at the glomerulus. [1]

(b) Calculate how many times more concentrated urea is in the urine than in the filtrate. [1]

(c) Explain why the concentration of urea in the urine is much higher than in the filtrate. [2]

(d) In one day, the kidneys of a healthy adult form 180 $dm^3$ of filtrate and produce 1.5 $dm^3$ of urine.

Assume that the filtrate and the urine are both made almost entirely of water.

Calculate the percentage of the water in the filtrate that is reabsorbed into the blood. Give your answer to three significant figures. [2]

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## 17 · Part 1 Supplement Excretion in humans [6]

Scientists measure the mass of protein eaten per day and the mass of urea in the urine per day for four adults. The adults all have the same body mass and do the same amount of exercise.

| mass of protein eaten per day / g | mass of urea in urine per day / g |
|---|---|
| 40 | 12 |
| 80 | 25 |
| 120 | 37 |
| 160 | 50 |

(a) Describe the relationship shown by the data. [2]

(b) Explain why eating more protein causes more urea to be produced. [3]

(c) Explain why urea must be excreted and not kept in the body. [1]

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## 18 · Part 1 Supplement Excretion in humans [6]

Two hours after a meal rich in protein, samples of blood are taken from two blood vessels:

- the blood vessel that carries blood from the small intestine to the liver
- the blood vessel that carries blood away from the liver.

The table shows the concentrations of amino acids and urea in the two samples.

| substance | blood going to the liver from the small intestine / arbitrary units | blood leaving the liver / arbitrary units |
|---|---|---|
| amino acids | 8.0 | 3.0 |
| urea | 0.3 | 0.9 |

(a) Calculate the percentage decrease in the concentration of amino acids as the blood passes through the liver. [2]

(b) Describe two ways in which the liver uses amino acids that would cause this decrease. [2]

(c) Explain why the concentration of urea is higher in the blood leaving the liver. [2]

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## 19 · Part 1 Supplement Excretion in humans [6]

One student writes six statements about excretion in humans.

1. Carbon dioxide is excreted through the lungs.
2. Urea is formed in the kidneys from excess amino acids.
3. Urine is formed from the liquid left at the end of the nephron after reabsorption.
4. None of the ions in the filtrate are reabsorbed into the blood.
5. Only a small amount of the water in the filtrate is reabsorbed into the blood.
6. Urea must be excreted because it is toxic.

Three of the statements contain an error.

(a) Give the numbers of the three statements that contain an error. [3]

(b) For each statement you gave in (a), write a corrected version of the statement. [3]

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## 20 · Part 1 Supplement Excretion in humans [7]

In a healthy adult, the glomeruli in both kidneys together form 125 $cm^3$ of filtrate every minute.

(a) Calculate the volume of filtrate formed in one hour. Give your answer in $dm^3$. [2]

(b) The concentration of glucose in the filtrate is 0.9 g per $dm^3$. Use your answer to (a) to calculate the mass of glucose filtered in one hour. [1]

(c) One patient has damaged glomeruli. The glomeruli in both kidneys of this patient together form only 30 $cm^3$ of filtrate every minute.

Calculate the percentage decrease in the volume of filtrate formed per minute, compared with the healthy adult. [2]

(d) The concentration of urea in the blood of this patient is much higher than normal. Explain why. [2]

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